

أثبت معادلة بور الآتية

$$\frac{1}{\lambda} = R_u \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

الحل: من العلاقة

$$\Delta E = E_i - E_f = hf = \frac{hc}{\lambda}$$

ولكن

$$E = \frac{E_1}{n^2}$$

$$\Delta E = \frac{hc}{\lambda} = \frac{E_1}{n_i^2} - \frac{E_1}{n_f^2} = E_1 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$$\frac{1}{\lambda} = \frac{E_1}{hc} \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

ولكن $\left(R_u = \frac{E_1}{hc} \right)$ إذا

$$\frac{1}{\lambda} = R_u \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$